

SYSTEMS AND METHODS FOR LONGITUDINAL PENILE PHYSIOLOGY MONITORING USING MULTI-MODAL RIGIDITY SENSING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to and is a continuation-in-part of prior U.S. provisional applications directed to distributed physiologic sensing platforms comprising wearable and insertable sensing nodes operating individually or cooperatively within a computational analysis system. The longitudinal penile physiology monitoring device described herein may operate independently or as a functional node within a multi-device biometric ecosystem for longitudinal physiologic analysis, adaptive calibration, compatibility modeling, or personalized physiologic profiling. All related disclosures are incorporated herein by reference.

FIELD OF THE INVENTION

The present disclosure relates to wearable biometric monitoring systems and physiologic sensing technologies. More specifically, the disclosure relates to systems and methods for longitudinal monitoring of penile physiology using a ring-form wearable configured to measure rigidity, tumescence dynamics, vascular behavior, thermal context, and motion-associated physiologic conditions across extended periods.

In various embodiments, the device comprises a body-positioned wearable configured as a closed ring, partially open ring, C-shaped structure, adjustable loop, deformable band, segmented wearable, or other geometry capable of maintaining positional association with penile anatomy during activity, rest, sleep, and episodic physiologic activation. The device may employ rigid, flexible, or hybrid material constructions and may be manufactured in multiple sizes or adaptive fit configurations to accommodate anatomical variability and comfort during soft-erect or erect conditions.

Rigidity and erection state may be inferred using one or more sensing modalities including strain sensing, pressure sensing, piezoelectric sensing, magnetic field sensing utilizing Hall effect and magnet arrangements, capacitive sensing, resistive sensing, optical perfusion sensing, temperature sensing, inertial measurement sensing, impedance sensing, acoustic sensing, or combinations thereof. Sensor configurations may range from simplified cost-optimized embodiments to higher-fidelity multi-sensor architectures.

Temperature sensing elements may provide physiologic context and environmental compensation, while inertial measurement units may enable posture detection, motion artifact rejection, activity classification, sleep-state segmentation, and optional movement telemetry characterization. Optional photoplethysmography sensing may capture pulse and perfusion-related physiological metrics.

Data generated by the device may be processed locally, transmitted to companion computing devices, or analyzed within remote computational environments configured to derive longitudinal physiologic profiles including erection onset speed, duration metrics, detumescence behavior, nocturnal erection characterization, stability indices, variability trends, and recovery patterns. The embodiments described herein are illustrative and non-limiting and encompass future sensing technologies capable of achieving substantially similar functional outcomes.